Fiscal Year:	FY 2020	**Task Last Updated:**	FY 04/03/2020
PI Name:	Vimal, Vivekanand Ph.D.		
Project Title:	Predicting Individual Differences in Learning to Manually Stabilize Attitude in a Space Flight Analog Environment		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	vvimaldhye@gmail.com	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	781-861-9697
Organization Name:	Brandeis University		
PI Address 1:	Ashton Graybiel Spatial Orientation Laboratory		
PI Address 2:	415 South St		
PI Web Page:			
City:	Waltham	**State:**	MA
Zip Code:	02453-2728	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 TRI-RFA-17-01: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
Start Date:	12/01/2017	**End Date:**	11/30/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Period of performance changed at some point to 12/10/2017-11/30/2020 per TRISH submission in March 2020 (originally was 1/1/2018-12/31/2019) (Ed., 4/2/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	DiZio, Paul Ph.D. (MENTOR: Brandeis University)		
Grant/Contract No.:	NNX16AO69A-P0202		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP In Specific Aim 1, to characterize individual differences in subjects, we will use three tests of increasing complexity. The first test, in collaboration with Dr. Faisal Karmali measures each subject's vestibular thresholds. The second test measures each subject's spatial perception threshold by finding where a subject's ability to indicate the balance point deteriorates when exposed to different magnitudes of relevant gravitational cues. In the third test, in collaboration with Dr. Mark Shelhamer, we will examine performance in a manual tracking task which has been able to predict motor adaptation ability in a different paradigm. Next, we will determine whether any of these tests predict learning and performance in a space flight analog environment. In Aim 2, to study learning, generalization, and transfer of skill we develop two training programs to help subjects improve performance in our space flight analog. One training program examines incremental exposure to a condition without relevant gravitational cues and the other exploits our recent finding that balance control consists of two dissociable processes: identifying the direction of dynamic balance and identifying the direction of gravity. We study retention by asking subjects to come back a month later and perform the same dynamic stabilization task. These specific aims address NASA roadmap gaps: SM24 (Determine if the individual capacity to produce adaptive change (rate and extent) in sensorimotor function to transitions in gravitational environments can be predicted with preflight tests of sensorimotor adaptability) and Train-04 (We do not know the types of skills and knowledge that can be retained and generalized across tasks for a given mission to maximize crew performance).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In this project we developed a machine learning technique that can, without prior human labels, cluster individuals into different groups that can be used to find suboptimal strategies and predict final performance. We created an effective training program that enhanced performance and learning in a dynamic stabilization task during a space flight analog condition. This training program confirmed a prior hypothesis that balance control consists of two dissociable processes, and showed additionally that each of which can be trained independently. Our training program further showed that improvements in performance and learning can be obtained even in a space flight analog task where a majority of subjects report spatial disorientation and where they cannot rely on gravitational cues to do the task. Moreover, this training can be retained for at least four months. Combined together, these findings can provide value to astronaut training.
Task Progress:	[Ed. note April 2020: Report submitted by TRISH to Task Book in March 2020; covers reporting as of November 2019.] Long duration space flights, such as to Mars, will involve multiple gravitational transitions that will result in many sensorimotor related difficulties that could jeopardize the mission. Our previous work shows that when subjects are deprived of gravitationally dependent vestibular and somatosensory cues, such as in low-g, 0g, and space flight analog environments, they easily become spatially disoriented and show poor learning and performance in an attitude stabilization task. In these experiments we secured subjects into a Multi-Axis Rotation System (MARS) device that was programmed to behave like an inverted pendulum, and participants were instructed to use an attached joystick to stabilize around the balance point. We created space flight analog conditions by having subjects dynamically balance in the Horizontal Roll Plane and in the Vertical Yaw Axis, where they did not tilt relative to the gravitational vertical and therefore could not use gravitational cues to determine their position and had to rely only on motion cues. In the first specific aim, our objective was to create an effective training program that would enhance performance and learning in our space flight condition. Our training program was based on our previous finding that balance control consists of two dissociable processes: aligning with the gravitational vertical using graviceptive position cues and dynamic stabilization using motion cues. Because subjects in space flight conditions can only rely on motion cues, our training program focused on refining subjects' abilities to dynamically stabilize independent of aligning with the gravitational vertical. We did this by setting the balance point to be different from the gravitational vertical in the Vertical Roll Plane. Subjects were instructed to search and then dynamically balance around this balance point. They were not told the location of the balance point and had to ignore biasing positional information from the gravitational cues and instead rely on motion cues to determine the location of the balance point. We found that subjects in this training group, who were later tested in our space flight analog condition (Horizontal Roll Plane) showed significant transfer of learning across many metrics whereas the control group, who balanced in the Horizontal Roll Plane without any training, showed very minimal learning transfer. We also found that the training group retained this learning when retested 4 months later. Subjects that received the training through a visual simulation did not acquire any enhancement in performance when tested in the Horizontal Roll Plane. These findings address NASA roadmap gaps: SM24 and Train-04. In the second specific aim, our objective was to characterize and predict individual differences with our space flight analog task using machine learning. We used a Bayesian Gaussian Mixture method to cluster subjects into 3 statistically distinct groups that represent Proficient, Somewhat Proficient, and Not Proficient performance. We found that subjects in the Not Proficient group exhibited a suboptimal strategy of using very stereotyped large magnitude joystick deflections. We also used a Gaussian Naive Bayes method to create predictive classifiers. As early as the second block of experimentation (out of 10), we could predict a subject's final group with 80% accuracy. Combined with the first specific aim, we now have the ability to characterize and predict individual performance in a space flight analog task, and then use an effective training program to enhance learning and performance in those that display poor performance.
Bibliography Type:	Description: (Last Updated: 06/09/2025)
Articles in Peer-reviewed Journals	Vimal VP, DiZio P, Lackner JR. "Learning and long-term retention of dynamic self-stabilization skills." Exp Brain Res. 2019 Nov;237(11):2775-87. Epub 2019 Aug 23. https://doi.org/10.1007/s00221-019-05631-x ; PubMed PMID: 31444539 , Nov-2019
Awards	Vimal V. "1st place in NASA HRP conference poster competition, 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019." Jan-2019